



Survival Rates of Unrestorable Fracture of Endodontically Treated Anterior Teeth Restored with Resin Composites or Crowns: A Retrospective Cohort Study

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Introduction

Unrestorable fracture is a common failure in endodontically treated teeth (ETT) [1]. ETT are susceptible to fracture due to extensive loss of tooth structure from dental caries, coronal fracture or pre-existing large restorations [2]. Direct resin composite (DRC) or full-coverage crown (FCC) is placed as a coronal restoration that reinforces ETT to withstand occlusal forces and protect from fracture [3, 4].

From a recent systematic review, anterior ETT with minimal to moderate loss of tooth structure could be conservatively restored with DRC to achieve an acceptable resistance to fracture; FCC would be required only if a significant loss of tooth structure is present [5]. The location of remaining cervical tooth structure is important for resistance to fracture [6]. For instance, the fracture strength of anterior ETT with partial loss of cervical structure and existing palatal wall is considered comparable with ETT having sound cervical structure [6].

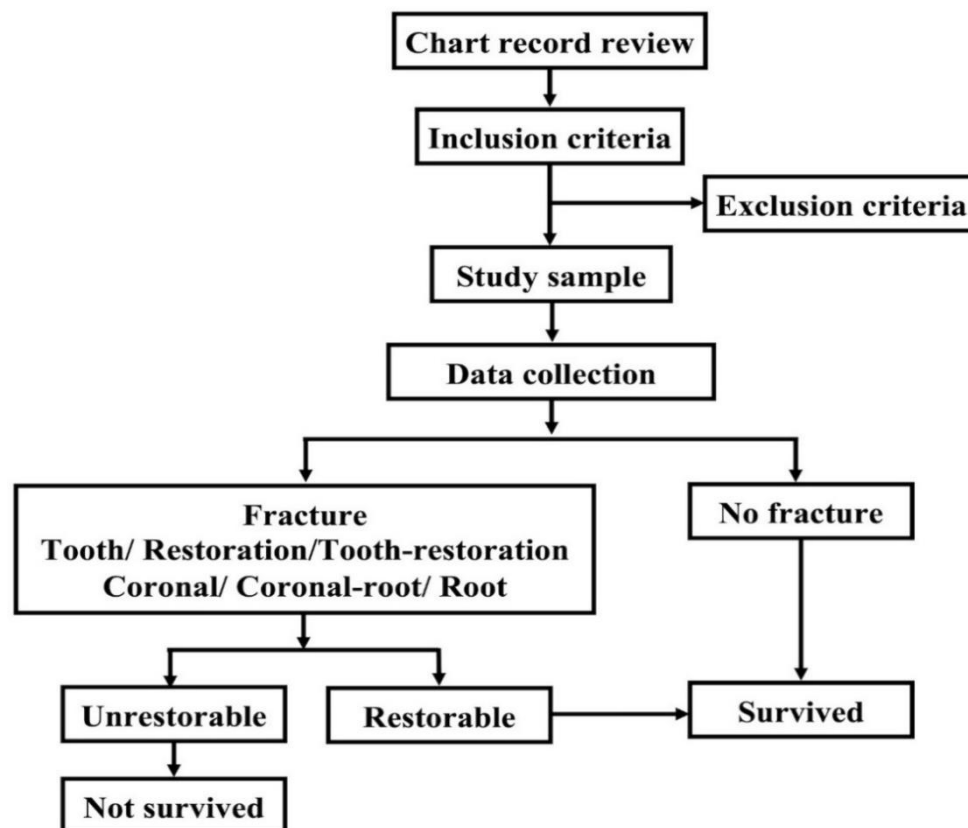


Figure 1. An overview of the methodology in this retrospective cohort study

Current clinical evidence about the effect of different coronal restorations on the survival from fracture of anterior ETT is scarce. From the recent clinical studies [7, 8], the benefit of FCC over DRC in prevention of tooth fracture is controversial. In addition, the reported clinical outcomes were combined from the data of anterior and posterior teeth. However, the risk of fracture in the anterior and posterior teeth are not the same and should be separately investigated. Only a retrospective study by Sorensen and Martinoff in 1984 [9] reported the separated data of anterior and posterior ETT. The survival rate from fracture of anterior ETT restored with DRC was higher than 90% and not significantly different from those restored with FCC.

Predisposing factors to unrestorable fracture have been reported in clinical studies of posterior ETT, *i.e.* remaining tooth structure, loss of adjacent tooth, or non-cuspal coverage restoration [10, 11]. From a clinical study in anterior ETT restored with glass-fiber posts and crowns [12], the survival rate of ETT with 3-4 remaining coronal walls tended to be higher, but was not significantly different from those with 1-2 remaining coronal walls. No other clinical study in anterior ETT has identified predisposing factors or directly compared between ETT with DRC and FCC restorations.

Current guidelines for restoration of anterior ETT are primarily based on laboratory results and insufficient clinical data. Therefore, this retrospective cohort study aimed to compare the survival rates from unrestorable fracture of anterior ETT restored with DRC and FCC. In addition, any predisposing factors to unrestorable fracture were identified.

Materials and Methods

The study protocol was approved by the institutional ethic committee (MU-DT/PY-IRB 2018/014.2302). Dental records and radiographs of anterior ETT were collected from dental charts of the patients who received endodontic treatments and attended recalls during the period of 2007–2019 at the Endodontic Postgraduate Clinic, Faculty of Dentistry, Mahidol University, Bangkok, Thailand.

Inclusion and exclusion criteria

Anterior ETT with DRC or FCC coronal restorations were included according to the inclusion and exclusion criteria. The inclusion criteria of ETT were (1) having mature root; (2) restored with single-unit coronal restorations, DRC or FCC; (3) having occluding pair; and (4) at least one-year recall period.

The exclusion criteria of ETT were (1) teeth with preoperative root resorption, crack, or fracture; (2) restored with direct or indirect veneers; (3) had supporting alveolar bone loss more than half of root length; (4) history of post removal before retreatment; (5) with procedural error(s) during endodontic and/or restorative treatment (such as root perforation); and (6) ongoing orthodontic treatment.

Endodontic and restorative procedures

Endodontic treatment was carried out under rubber dam isolation. Conservative access and working length determination were performed. Root canals were prepared using the crown-down technique using hand and/or rotary Ni-Ti files, and irrigated with 2.5% sodium hypochlorite and 17% EDTA. All root canals were medicated with calcium hydroxide paste for one to four weeks. The root canals were obturated with gutta-percha and root canal sealer [zinc oxide eugenol or epoxy resin sealer (AH-Plus, Dentsply-Maillefer, Tulsa, OK, USA)], using lateral or warm vertical compaction, up to 1-2 mm level below the cemento-enamel junction (CEJ). The teeth were planned for coronal restorations, DRC or FCC.

DRC restorations were placed as a permanent restoration, except in the cases that intermediate restorations were required prior to FCC. The access cavities were lined with non-eugenol, zinc-oxide based material (Cavition, GC corp., Tokyo, Japan) and/or GIC lining (Vitrebond, 3M ESPE, St. Paul, MN, USA; or Fuji VII, GC corp., Tokyo, Japan) at 1-2 mm thick. The cavities were bonded with a resin-based adhesive [etch and rinse adhesive, Adper Single Bond 2 (3M ESPE, St. Paul, USA); or self-etch adhesive, Clearfil SE Bond (Kuraray, Osaka, Japan)] according to the manufacturers' instructions. Micro-hybrid or nano-hybrid resin composite [Z350 XT (3M ESPE, St Paul, USA), or Estelite Sigma (Tokuyama Dental Corp., Tokyo,

Japan)] was filled into the cavities using the incremental technique in 2-mm thick and light cured for 20 sec in each layer.

FCC restorations with a placement of intra-radicular post was planned. Post spaces were prepared using electrical heat carriers (System B, SybronEndo, Orange, CA, USA) or Peeso-reamer drills (Dentsply, Maillefer, Tulsa, OK, USA). Cast metal posts or prefabricated fiber posts [D.T. Light-Post (BISCO Inc., Schaumburg, IL, USA)] were prepared and cemented into the canals with a resin-based material [Rely-X U200 (3M ESPE, USA) or MultiCore Flow (Ivoclar Vivadent AG, Amherst, NY, USA)]. All-ceramic crowns (Empress E. Max, Ivoclar Vivadent AG, Schaan, Liechtenstein) or porcelain-fused-to-metal crowns were fabricated and cemented on the prepared teeth.

Predisposing factors and incidence of fracture

An overview of the methodology is present in [Figure 1](#). The information were collected from dental records and radiographs. The age, gender, and recall periods were noted. The details of preoperative, intraoperative, and postoperative factors were identified as follows: (1) tooth type (incisor, canine) and location (maxillary, mandible); (2) type of coronal restorations (DRC, FCC); (3) surface(s) of restoration for DRC, or core build-up for FCC; (4) incidence of tooth/root fracture (yes, no); (5) fracture site (coronal, coronal-root, or root); (6) posterior support, with at least one pair of occluding teeth (none, 1, or 2); (7) adjacent teeth (none, 1, or 2); (8) type of post (none, cast metal, or prefabricated fiber post); (9) relationship between post length and alveolar bone level (below, above, or at the level); (10) wall(s) of remaining cervical tooth structure; (11) cervical root dentin thickness (mesial and distal sites) at 2 mm below the CEJ, calculated into a ratio to the root canal width ($< 1:1:1$ or $\geq 1:1:1$); and (12) parafunctional habits, such as bruxism or clenching (yes, no).

Table 1. Survival rates and statistical analysis of anterior ETT according to the variables ($n=263$)

Factors	Number of teeth (n)	Survival (%)	Not survival (%)	P-value
Gender				
Female	155	147 (94.8)	8 (5.2)	0.59
Male	108	100 (92.6)	8 (7.4)	
Tooth type				
Incisor	185	176 (95.1)	9 (4.9)	0.13
Canine	78	71 (91)	7 (9)	
Location				
Maxillary	204	191 (93.6)	13 (6.4)	0.45
Mandible	59	56 (94.9)	3 (5.1)	
Restoration				
Composite	157	142 (90.4)	15 (9.6)	$<0.01^*$
Crown	106	105 (99.1)	1 (0.9)	

* A significant difference from the log-rank test ($P<0.05$)

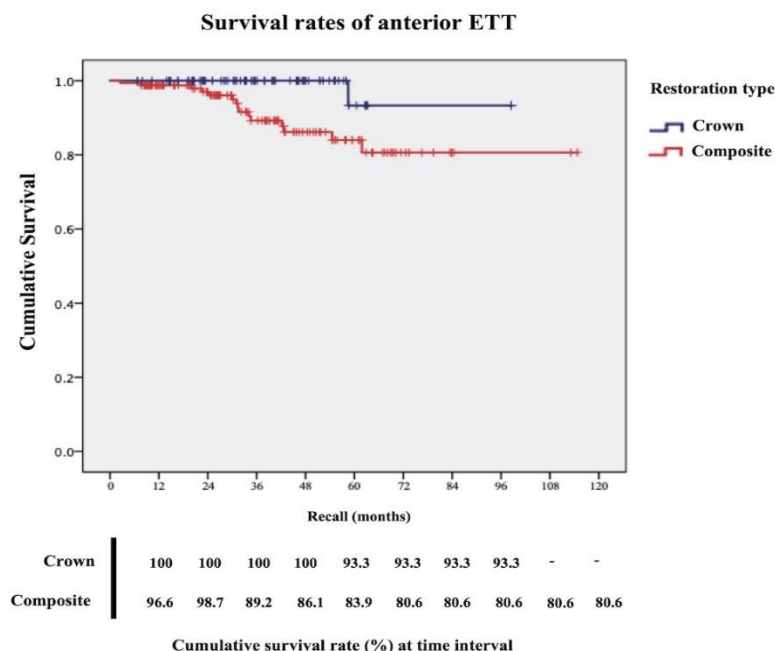


Figure 2. Kaplan-Meier survival analysis from unrestorable fracture of anterior ETT with resin composite and crown restorations, and the cumulative survival rates at the time intervals

Restorative surface(s) of resin composite or core build-up were noted in mesial (M), distal (D), palatal (Pa), labial (La), and incisal (I) surfaces, which were further categorized into two groups as 1-3 or 4-5 restorative surfaces.

Remaining cervical tooth structure was identified according to the residual walls at the cervical area in M, D, Pa, and La. The cervical tooth structures were further categorized into two groups: 0-2 and 3-4 walls. For the maxillary anterior ETT, the cervical tooth structures were subcategorized, according to the remaining wall of cervical tooth structure, into the four subgroups- (1) 'complete', all four walls existed; (2) 'none', all walls were absent; (3) 'partial with Pa', other wall(s) were missing but the palatal wall remained; and (4) 'partial without Pa', the palatal wall was missing but the other wall(s) remained.

A ratio of root dentin and root canal width at the cervical region (2 mm below the CEJ) was estimated. Root dentin thicknesses (M and D) and root canal width (C) were measured from the post-operative radiographs. M:C:D ratio was calculated and classified into two groups, $< 1:1:1$ and $\geq 1:1:1$.

Incidence of fracture was noted into restorable and unrestorable. The definition of restorable fracture was the fracture that can be repaired or replaced with a new restoration. The definition of unrestorable fracture was the fracture with an extensive destruction of tooth structure leading to tooth extraction. The survival rate from unrestorable fracture was calculated into percentage.

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Conflict of Interest: 'None declared'.

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